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Prediction of Groundwater Potential Mapping using Linear Regression and Genetic Algorithm

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Abstract

Groundwater capacity studies assess subsurface resources using multiple methodologies. Groundwater mapping potential estimates can help manage and schedule resources. Finding good groundwater development locations is essential for sustainable management. Geological, hydrological, and environmental data are needed to map underground water potential. Water resource management and hydrogeology can influence groundwater mapping potential. Several methods have been used to predict groundwater potential mapping. This study estimates groundwater potential using genetic algorithms and linear regression. The recommended linear regression approach estimates groundwater potential and parameters. Select the most critical groundwater potential variables and optimize the linear-regression model. Groundwater mapping potential is predicted after model optimization. An area with different hydrogeological parameters was examined to test the approach. The proposed method accurately predicts groundwater potential mapping and provides relevant water resource management data. LR-GA helps manage water assets and calculate groundwater potential. The advice is 93.7% correct. The proposed approach is 90.6% sensitive and 89.63% specific.

Keywords: Linear regression; Genetic algorithm; Prediction; Groundwater Potential; Thiruvallur;

I INTRODUCTION

The primary supply of clean drinking water worldwide is groundwater, which also significantly affects housing needs and agricultural needs [1]. Before a few decades, population growth, urbanization, improper chemical manure use, climate variation, and weak water source administration caused groundwater superiority to collapse in various locations, including Thiruvallur district in India. Constant ejection from the ground depletes groundwater. By understanding the insufficiency of water all over the lands, people managed to drink the water they pursue because there is no source to get water. Therefore, some improvements are needed in the areas that are marked under water insufficiency

[2].Groundwater corruption hinders human expansion, therefore experts' study underground waterway evolution. Additionally, it aids decision-makers in comprehending how to safeguard the quality as well as the scope of the assets. [3]. Fresh water fountain between plant canopy and aquifer is dangerous. The dangerous zone was carefully examined to understand groundwater flow's role, structure, and process. Petrology, morphology, mud, drainage thickness, feature concentration, slope, precipitation, land-use. outline are taken under consideration to label the areas where the groundwater is possible [4]. Therefore, the final groundwater potential mapping is formed when all these factors have been synchronized. Petrology, morphology, mud, drainage thickness, feature concentration, slope, precipitation, and land use determine groundwater potential. Together, such traits create the ultimate potential mapping. Many studies sought groundwater spatial ambiguity reduction. Although remote sensing gives useful data, it is rarely employed to track environmental concerns. Even if hydro doesn't exist on the outermost layer, sensors on the ground can inadvertently determine it [5]. Thus, slope maps calculate groundwater potential zone difficulties. The groundwater incursion and surface overflow are affected soon. Low level intrusion has heavy intrusion and less surface overflow whereas, high level intrusion possesses low intrusion and heavy surface overflow [6]

Various groundwater dimensional methodologies are used in the previous studies [7]. Hydrological performance, field hydraulic leading, and investigative piercing are time-consuming and expensive groundwater dimension methods. But they are also partial to small scale dense data captured from peak calculations. Organic research requires genetic algorithm and linear regression. LR/GA repair methods. For its simple modeling of groundwater

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potential and affecting variables, linear regression provides a clear, interpretable baseline. Groundwater systems are complicated and nonlinear, hence the genetic algorithm is vital. Optimizing regression model parameters using an iterative, evolutionary method improves model performance for nonlinear groundwater potential. Combining linear models' simplicity and evolutionary algorithms' adaptability improves forecast accuracy and reliability. LR probability is 0–1. LR analyzes 12 environmental features. Validation employs six AUC criteria. If the dependent variable is under the binary level, the binary LR analysis is utilized to calculate whether the factor is present or absent considering the independent variable [9]. Optimizing regression model parameters using the genetic algorithm improves model accuracy. Although linear regression models linear correlations between groundwater potential and influencing variables, it may struggle with complicated, nonlinear groundwater systems. The genetic algorithm simulates natural selection by setting model parameters through selection, crossover, and mutation. This evolutionary technique lets the model explore more possibilities and find the best parameters to capture both linear and nonlinear patterns, boosting groundwater potential mapping forecast accuracy and resilience. [11]. A GA has 0s and 1. Every expansion from randomly established personalities has a group. Assessing generational population suitability. Suitability impacts optimization [12]. Random mutations, genetic recombination, and stochastic selection generate generations. After population reaches fitness or max groups. Data is deleted and traits extracted by genetic algorithm. Optimization issues disappear [13].The key contribution of the developed framework is summarized as:

1. Mapping the groundwater potential zones in the Thiruvallur region is the primary objective of this work. We will achieve this by combining genetic algorithmic
2. Compared accuracy with existing approaches to establish that the proposed method performs effectively in ground water prediction

Part 2 adds content and compares methods. Sections 3 assess the topic, 4 present the LR-GA technique, 5 create evaluation criteria, and 6 give outcomes and future study.

II Related Works

Groundwater supports water resources, say Banadkooki et al. [14]. This paper forecasted groundwater levels using three models: Second model: MLP/RBF/GP. Several math models predicted groundwater. Researchers mapped ANN groundwater potential. High precision and performance. These models fix healthy ANN models, recognize ANN-optimization process linkages, and understand input functionalities' ambiguity. Water theft and corruption cut supply. MLP wavelet has less virtual error than RBP/GP. Delay in convergence is this paper's greatest fault. The paper reveals that the management of groundwater and remediation are benefited from the modeling-optimization models of the groundwater and the impurity measure said by Seyedpour et al [15]. Multi-quadratic Radial Basis Function links groundwater flow, sensitive pollution, and oxidant transport in Meshfree.

Enhance models by adjusting hyperparameters, employing cross-validation, and utilizing regularization approaches. Investigate advanced methodologies such as hybrid models and ensemble methods, and contemplate the utilization of more sophisticated evolutionary algorithms. In two steps, groundwater potential is predicted. Adaptive Neuro Fuzzy Inference System (ANFIS) advanced phases predicted groundwater extraction optimization. using fall, last month's surface water volume, two months later, and current water source. Bad performance for data testing.Optimizations include PSO, GWO, HHO. To learn ANFIS. It improves precision. ANFIS method HHO-ANFIS is optimal, per Taylor diagram.Kini et al.[17]

This work trains DE, ANFIS, PSO, ACO, and CGA groundwater quality variables. Using parallel and advanced investigations, we validate Total Hardness (TH), Sodium Adsorption Ratio (SAR), and Electrical Conductivity (EC) via inputs Training superiority variable quantity follows in ANFIS. This method is efficient by R2, Complex analysis, RMSE, and MAPE. EC/TH enhancements help all algorithms. ANFIS has several benefits, however its training is ineffective.Report by Yadav et al.[18] He says history Water is scarce in India. By 2020, 50% of large nations will have groundwater issues. Corruption, urbanization, population increase, and weather change deplete water. Previously, machine learning (ML)-based Singular Spectrum Analysis (SSA), Mutual Information (MI), Genetic Algorithm (GA), Artificial Neural Network (ANN), and Support Vector Machine (SVM) methods solved this problem Use NOI, SOI, NINO3, groundwater, rainfall, weather, and monthly world population projections. Preprocessing improves MI, GA, and delay breakdown tests. Recent groundwater mapping utilized SVM/ANN. Pipeline scarcity leaks largely affect physical modeling.Due to rising demand, Sepahvand et al.[19] Weather affects water. For groundwater management, this work applies a reproduction optimization model bi-objective non-dominated sorting genetic algorithm-Type II. It reduces irrigation water shortages and enhances core crop net revenues. Genetic groundwater

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forecasting. Module syncs MOGA optimization. To study different climates, drizzle, dehydration, and forecasting periods were introduced. The latest analysis shows higher net profitability. This technique typically fails summer aquatic needs.Wei etal.[20]Genetic algorithms GA-BPNN and GA-SVM. The hybrid design improves GA-SVM over GA-BPNN. The slope's groundwater component was simulated numerically. SPSS calculated effect time using Pearson correlation. GLF negative, GLFD ignored, serious worry.AlAyyash et al. predicted groundwater[21] Hydrogeology may make SINTACS ignore groundwater dangers.Panahi say recent water shortages are the biggest issue et al.[23].Machines and deep learning map groundwater. Both CNN and SVM predict mapping. Functions estimate much. Data is collected for testing and training. The dataset is 70% training, 30% validation. GCFs, Geology, Daytime anisotropic Heating, Convergence Index, Slope Height, Convexity, Catchment Area, Slope Angle, TOPO, Valley Depth, LS, MBI, TX, TRI, LC, and FP measure slope, ground test SVR/ROC-AUC CNN CNN predicts groundwater better than SVR. Not comprehending the method's foundation is its worst shortcoming.Mosavi [24]. Groundwater is dirtier and more disturbed than surface water, study reveals. Fixing them takes time and money. With prediction models and preemptive actions, delegates control groundwater. Machine learning (ML) hardness susceptibility mapping approaches are useful for groundwater quality modeling but have received little attention. MDA examines BRT and RF. Problem: critical parameters alter hardness models.

III Problem Statement

Multivariate Discriminant Analysis (MDA) based on machine learning (ML) is the methodology utilized for forecasting the feasibility of mapping groundwater Mosavi et al.[24]. Restoring mapping potential takes time. Delegates make informed decisions and sustainable groundwater governance via prediction modeling and prevention. Detailing may slow hardness modeling. Genetic Algorithm and Linear Regression solved.

IV Proposed Methodology for Predicting Groundwater potential

The proposed ground water potential prediction method in various activities is depicted in Figure 1. Studies begin with Thiruvallur district data. Normalization, missing data imputation, and mutual information preprocess data. Linear regression calculates dependent and independent variables. Genetic algorithm improves regression. LR-GA predicts groundwater potential.

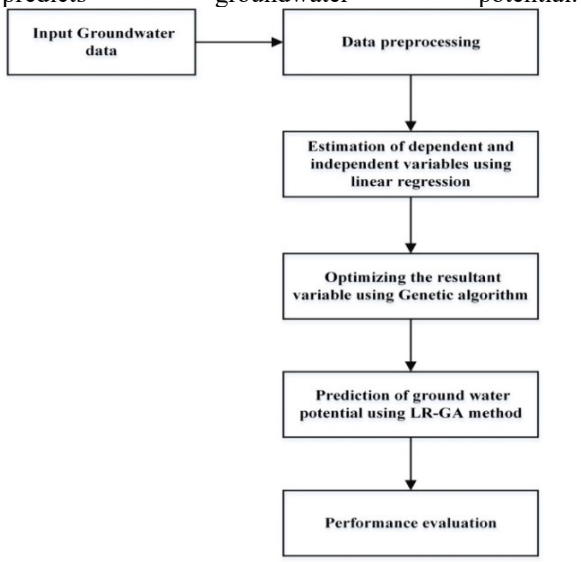


Figure 1: Work flow of the proposed method

A Database Collection

a Study area

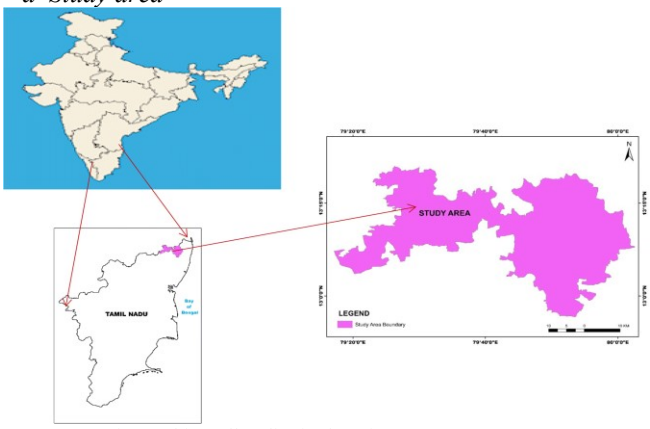


Fig 2: Thiruvallur district location map [25]

To determine groundwater potential, the research region was classified into extremely low, moderate, high, and very high levels. Groundwater accessibility depends on geology, topography, soil, precipitation, drainage, slope, physiography, and land use. Previous environmental and weather data is compared for the research region [26].Fig (2) represents the study conducted in thiruvallur district.

b Tropical climatic condition in Thiruvallur basin

Chennai-Thiripallur basin [26]. It has longitudes 79° 10′ 32″E and 80° 25′ 15″E and latitudes 12° 40′ 32″N and 13° 39′ 0″N. Andhra Pradesh, Kanchipuram, Vellore, Bay of Bengal, Chennai. Annual temperatures are 24.3°C–32.9°C in tropical zones. Tamil Nadu and Andhra Pradesh form the basin. The research region has a semi-arid tropical climate with temperatures between 13.9°C and 45°C, according to CGWB 2008. Thiruvallur has the highest summer temperatures and

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Tiruthani the lowest winter temperatures. Sunlight changes with seasons.

c Rainfall in the study area

1104 mm of rain falls annually in the studied region. Northeast and southwest monsoons rain 52% and 41%. The Thiruvallur basin averaged 775mm and 1580mm rain. 50%–80% humidity and 5.70–14.16 km/hr wind speed characterize the basin. Wet well water might reach 3–4 meters. Rivers Nandi, Koum, Koratalayar, Nagri, and Araniyar flow east-west through the basin. Backwaters and tides from the Gulf of Bengal affect the Araniyar and Korattalayar rivers. All rivers are transitory and have substantial rainy-season volumes, supporting dendritic drainage. West-east and north-south groundwater rotation.Rainfall in Thiruvallur district for 40 years is mentioned in Fig (3).

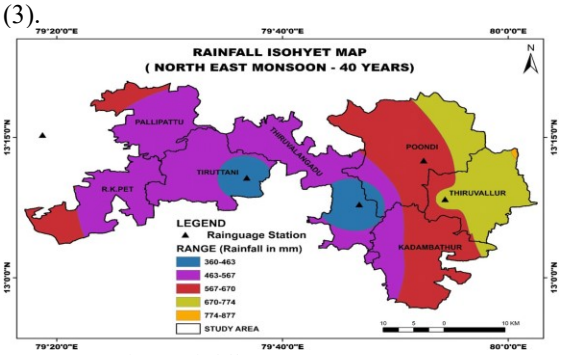


Fig 3: Rainfall map [25]

B Data preprocessing

[27] Prepare groundwater level forecast data to eliminate mistakes. Data cleansing, standardization, and transformation may be needed. A prediction uses groundwater, rainfall, temperature, drainage density, land use, lineament density, and geology. For rainfall and temperature analysis, time series are produced and missing data imputed. This is needed for all big data analytics and data mining projects. Statisticians substitute mean, median, or mode for missing numbers. Pre-processing includes missing value imputation and normalization after duplication removal. Starting with high-quality data on soil type, topography, rainfall, and land use is essential for linear regression and genetic algorithm groundwater potential mapping preparation. To ensure dataset integrity, impute or remove missing data. Use normalization or standardization to guarantee all characteristics contribute equally to linear regression. Use correlation or principal component analysis to pick the most important variables affecting groundwater potential. Check for outliers and modify variables to meet linear regression assumptions. This extensive preprocessing creates a clean dataset that improves model performance and prediction.

a Missing data imputation

When some variables in the dataset are missing values, the average of the ones that are there are used to fill them in. Here is the mathematical expression for that

$$O_i = \frac{O_{i-1}+O_{i+1}}{2}, \quad i \in N \tag{1}$$

In eq. (1), O_i N stands for the natural numbers, For each missing value, the corresponding operations are O_i, O_(i-1) for the preceding value, and O_(i+1) for the subsequent value. Despite the availability of numerous normalizing procedures, min-max normalization is utilized to normalize the data and fit it within a certain range.

Important aspects of standardization significant features is dependent on a wide range of variables, including environmental geology, terrain, weather, and human actions. f Normalization

1. Elements individually undergo scaling.
2. The size of the data set: large, medium, or tiny does not affect the normalization process.
3. The data size is not their responsibility.
4. The range of possible scaling values is from 0 to 1.
5. Only integers can be normalized.

. Equation for normalizing the variables is given below;

$$N = \frac{(|A|)-(10^{m-1})*(|C|)}{10^{m-1}} \tag{2}$$

The variables A, m, and N are data elements, the initial digit of A being C, and the scaled one value ranging from 0 to 1 being N. [28].

b. Mutual information

To choose the factors that are most important from the gathered data, the process of feature selection is employed. The quantities of information that a single random variable has about another one is measured as mutual information. The mutual information among the two unknown variables A and B determines the way it is computationally expressed in eq. (3)

$$I(A,B) = \iint dadb(x,y)log\frac{\mu(a,b)}{\mu_a(a)\mu_b(b)} \tag{3}$$

Where the marginal densities of A and B are $\mu_a(A) = \int db\mu(a,b)$ and $\mu_b(B) = \int da\mu(a,b)$. The mutual information between A and B is zero if and only if A and B are independent. If A and B are highly dependent, then the mutual information will be high, indicating that knowledge of one variable provides significant information about the other variable.To identify closely connected inputs and outcomes, information theory calculates marginal, conditional, and joint entropies. Positive correlation indicates strong relationship; zero indicates no association [29].

C Ground water influencing factors

Environmental geology, terrain, weather, and human behavior affect research region groundwater

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factor selection. Twelve groundwater mapping criteria are chosen via research. These include soil type, geology, elevation, slope, aspect, river density, rainfall, land use, and flow direction. Hydrological effects can be determined by classifying topography into nine groups [30].

Aspect: Geographical features, weather, and human behavior influence research region groundwater factor selection. Hydrological effects are determined by nine terrain classes [30]. First, collect and process geological, hydrological, and climatic data. Normalize and prepare data for modeling, including all required characteristics. Use linear regression to create a baseline model that forecasts groundwater potential from input factors. To capture nonlinear correlations in the data, use a genetic algorithm to enhance linear regression model parameters by imitating evolutionary processes including selection, crossover, and mutation. Finally, cross-validate and test the model on a distinct dataset before creating the groundwater potential map to confirm accuracy and generalizability.

Weather and height affect water runoff beyond terrain. Runoff and infiltration are affected by flow direction, therefore hydrological assessments should include it. Hydrologically, slope is crucial. Lower slopes increase surface water infiltration and groundwater recharge. Thiruvallur slope is on Fig. 4.

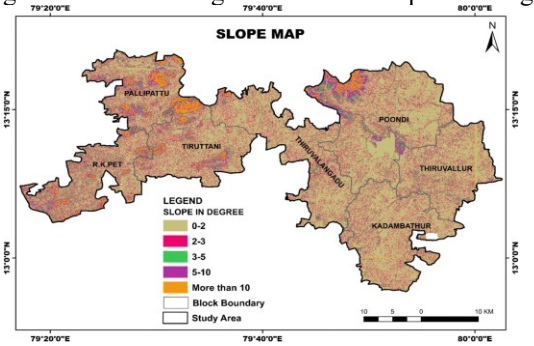


Fig 4: Slope map [25]

Land use: Human activity and land use alter groundwater potential. Concrete buildings may hinder replenishment owing to land use runoff. Deforestation inhibits soil water absorption. River density reduces infiltration and recharge, lowering groundwater potential. The research region has several rivers, thus runoff will be high and recharging low. LULC map of Thiruvallur (Figure 5).

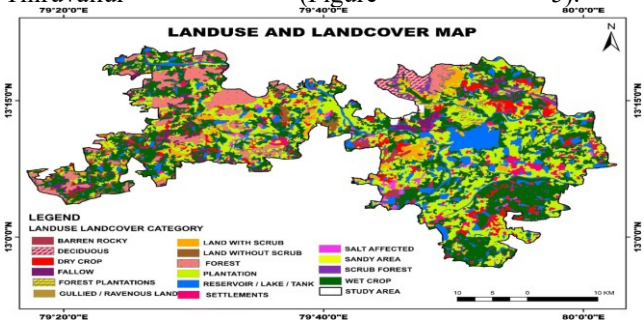


Fig 5: Land use and land cover map [25]

Drainage density: Water table depth and drainage density affect groundwater recharge. Also, high drainage density may cause erosion. Soil permeability and drainage vary by geology. The outermost subsurface water layer maps show aquifer recharge and outflow [30]. Geology, climate, and soil permeability determine drainage density—stream proximity—in the research location. Areas with little drainage are prioritized [31]. Thiruvallur drainage density.

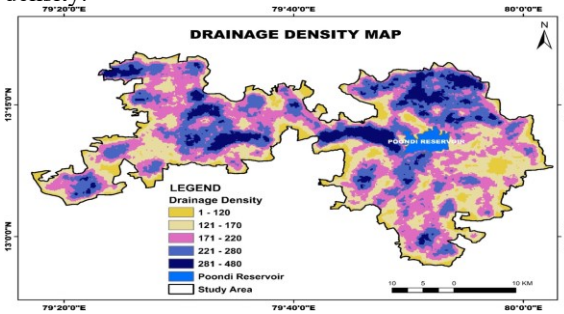


Fig 6: Drainage density map [25]

Lineament density: Satellite photography shows geological and non-geological fracture zones, joints, and drainage patterns as lines. Deformed bedrock with groundwater potential has fractures, faults, and joints. Hydrologically productive high-lineament density areas rank higher. Thiruvallur district lineament map.

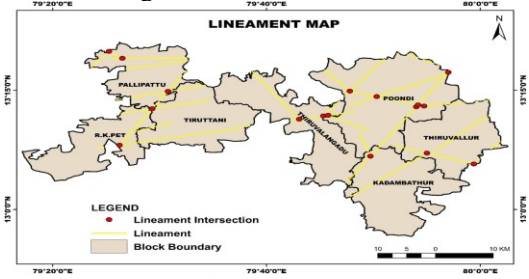


Fig 7: Lineament map [25]

D Linear regression analysis

Linear regression expresses the relationship between important factors that predict and then a dependent variable Given the dataset $\{y_j, x_{j1}, \dots, x_{jq}\}_{j=1}^{m}$ with q explanatory predictors $(x_{j1}\ to\ x_{jq})$ and measurements of groundwater

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recharges (y_j) containing n quantitative data. Thus, the LR approach can be derived using eq. (4).

$$g_0(x_j) = \theta_0 + \theta_1 x_{j1} + \cdots + \theta_p x_{jp} = x_i.\theta, j = 1, \ldots, m \quad (4)$$

Where, $g_0(x_j)$ is the computed resultant variable, x_j ($x_{i1}\ to\ x_{jq}$) is the descriptive predictors, θ_0 represents the intercept, and θ_q is slope coefficients for each descriptive predictors [32]. The main goal of MLR (independent variables) is to estimate the state of relationship between a response variable (dependent variable) and two or more predictors [33].Top of FormEquation (5) shows the generic form of an MLR mode

$$1.\hat{S} = x_0 + \sum_{i=1}^{n} x_i C_i \quad (5)$$

where $\hat{S}$ is the model‘s output, $C_i's$ constitute the model's independent input variables, and $x_0, x_1, x_2, \ldots, x_m$ are partial regression coefficients.

E Optimization using Genetic algorithm

Genetic algorithm optimizes linear regression. GA evolves genetically and naturally. It optimizes SEO and ML. Population initialization, fitness, mutations, crossover are GA algorithm phases. This approach precludes local optima and ensures convergence via crossover and mutation. GA selects top solutions to produce the next generation. It continues until satisfaction [34]. Linear regression was chosen because it is easy to understand and works well for finding linear relationships between groundwater potential and different factors that affect it. Genetic algorithms are used because they can solve difficult, complicated problems by mimicking natural selection processes. This makes them good for improving model parameters and making predictions more accurate. The study uses the best parts of both linear and nonlinear models to make a strong and accurate groundwater potential map by mixing these two methods.

F Prediction of Groundwater Potential using LR-GA

This study classified groundwater maps as extremely low, moderate, high, and low. Hydrogeological and geological parameters are used by the GPI to determine groundwater availability. Formulating groundwater potential index values utilizing Equation (6) and the "weighted overlay analysis tool,

$$.GPI = \sum_{w=1}^{m} \sum_{j=1}^{n} (W_j * X_i) \quad (6)$$

Where W_j indicates the normalized weight of the *j* thematic layer, X_i There are n theme characteristics, m theme levels, and it is the layer's its feature's normalized weight. A region's groundwater potential was evaluated extremely low, low, very high, high, and moderate by the GPI [35].

V RESULT AND DISCUSSION

Model performance is quantified by RMSE, R-squared, and MSE. The suggested approach's forecasts are compared against groundwater levels for accuracy. Equation (7) calculates MSE.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(t_i - a_i)^2 \quad (7)$$

Where, N is the amount of data, t_i the validation stage (check step) in data-driven modeling methods like ANN modeling is constructional and independent of training. The last test predicts machine algorithm accuracy. LR-GA evaluation requires model validation. Many studies used the ROC curve to objectively verify prediction models. The ROC curve validated the LR, GA, and LR-GA models in this study. We calculated LR, GA, and LR-GA model success rates using ROC curves. Proposed approach is 93.7% correct. The suggested method is 90.6% sensitive and 89.63% specific. Quality input datasets need rigorous data preparation, including feature selection, normalization, and missing data treatment. Avoid overfitting by optimizing the linear regression model with hyperparameters and regularization. Iteratively refine model parameters with the genetic algorithm to capture nonlinear interactions. Use cross-validation to evaluate and change model performance. Domain expertise and innovative methodologies like ensemble methods can improve model precision and groundwater potential forecasts.

Table 1: Spatial-variation of the ground water potential

x	Areas in percentage		
	LR	GA	LR-GA
0.0-0.2 (very low)	12.47	25.27	15.82
0.2-0.5 (low)	16.99	13.93	14.02
0.5-0.7 (moderate)	20.70	6.98	13.96
0.7-0.9 (high)	17.63	18.22	19.28
0.9-1 (very high)	32.21	36.60	36.92
Overall	100.00	100.00	100.00

Table 1 describes the spatial-variation of the ground water research used in the research study.

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Table 2: Comparison in terms of accuracy, sensitivity and specificity

Methods	Specificity	Sensitivity	Accuracy
LR	85.6	82.5	83.12
GA	85.7	82.1	83.89
Proposed LR-GA	89.63	90.6	93.7

Fig. (2) Shows Thiruvallur basin linear regression, genetic algorithm, and LR-GA geographic variation. The Thiruvallur district water basin proportion. River basins are predicted more commonly by proposed method. Success requires validating model predictions. Use kappa, specificity and accuracy, sensitivity, and ROC curve. A confusion matrix or ML cross-table has four outcomes: false positive, false negative, true positive, and true negative. The linear regression model's accuracy is verified by comparing predicted and observed groundwater recharge levels using the learning parameter matrix.

Accuracy, sensitivity and specificity comparison of proposed method with existing method is represented in table (2). The accuracy of ground water level prediction using LR-GA achieved a rate of 93.7%.

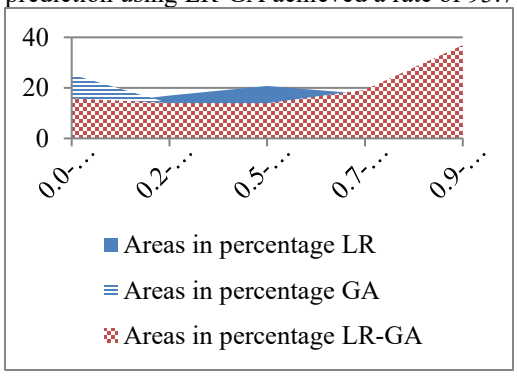


Fig 8: Ground water potential areas in terms of five classes

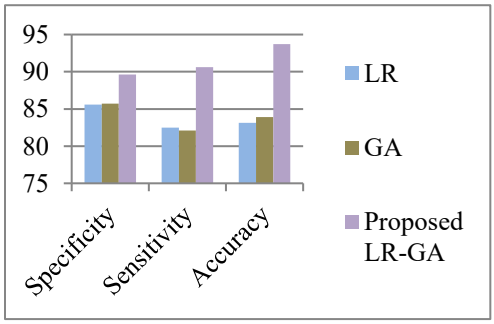


Fig 9: Comparison of performance metrics

In figure 9, comparative analysis is carried out separately for LR and GA. When LR-GA is combined, it exhibits highest accuracy than existing methods.

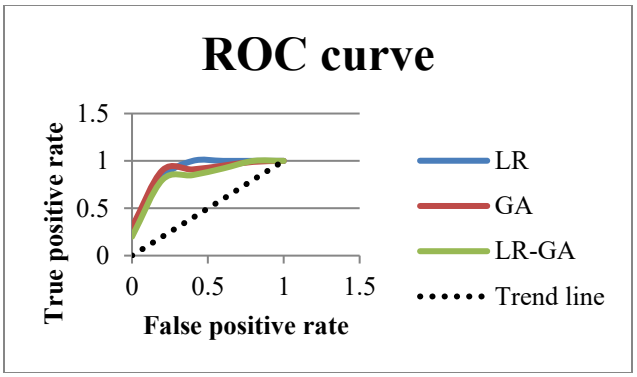


Fig 10: ROC curve for proposed and previous methods

In figure 10, ROC curve for LR, GA and LR-GA is represented and the trend line is plotted diagonally, which shows the proportionality of both the axes.

VI Conclusion and future work

The proposed method has been evaluated through a case study and has been found effective in predicting groundwater potential mapping. Therefore, it is determined that the suggested method is an effective strategy for forecasting groundwater potential mapping, which can provide a better tool for water resource administration and scheduling. The proposed method provides a foundation for future research in groundwater potential mapping. Further work can be done to optimize the proposed method by integrating additional data sources such as remote sensing data, groundwater level data, and socio-economic data. The use of advanced ML techniques such as RF, DT and SVM can also be explored. The suggested approach can be employed to different hydrogeological conditions and can be validated using independent datasets. The future work can also focus on developing an interactive web-based platform for groundwater potential mapping to facilitate decision-making for water resource management and planning.

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